Does Depth Help Nutrition Estimation Generalize to Unseen Foods?

CS7150 Deep Learning, Term Project Proposal

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1. What It Is

We test whether depth information helps a model estimate the nutrition of foods it has never seen in training. From one overhead photo of a dish, the model predicts five values: calories, mass, fat, carbs, and protein.

2. Why

The problem: photos hide portion size. From the top, a small and a large portion can look almost the same, yet differ greatly in calories. Depth shows how high the food is piled, which reveals the amount.

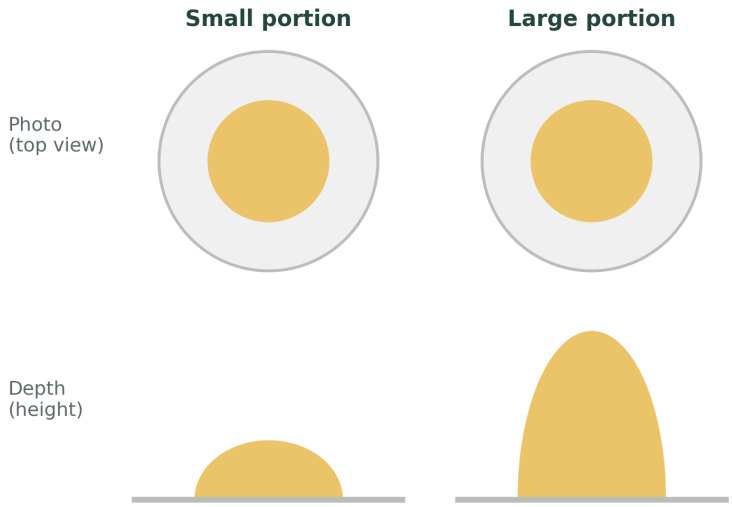


Figure 1. Same top view, different amount. Depth captures the height that a photo misses.

Why unseen foods matter. A real nutrition app will often meet dishes it was never trained on. A photo-only model may fail when it cannot recognize the food, while depth describes size regardless of food type.

3. Background and Existing Solutions

Prior work shows that depth improves nutrition estimation, but always on familiar foods.

Work	Depth source	Tests depth on unseen food types?
Nutrition5k dataset [1]	Depth camera	No
RGB-D fusion networks [2]	Depth camera	No
DPF-Nutrition [3]	Estimated from photo	No
Multimodal LLMs [4]	None	Not applicable
Ours	**Depth camera**	**Yes**

Gap: existing studies test on the same kinds of food used in training, so it is unknown whether depth still helps on new food types.

4. Research Questions

1. **RQ1:** Does depth help more on unseen foods than on seen foods?
2. **RQ2:** On unseen foods, is the error from portion size or from food type?
3. **RQ3 (optional):** Can depth estimated from a normal photo replace a depth camera?

5. Approach

5.1 Data

We use Nutrition5k, about 3,500 real cafeteria dishes. Each dish has a photo, a depth image, and weighed nutrition labels.

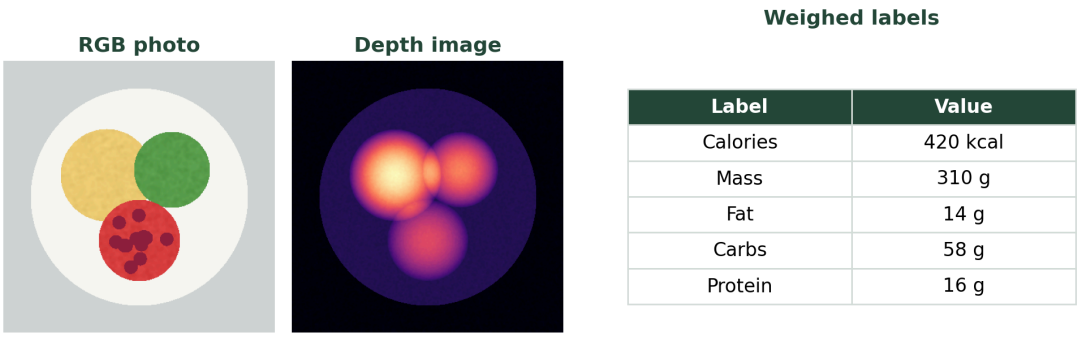


Figure 2. One Nutrition5k sample (illustrative): photo, depth image, and labels.

5.2 Seen vs. Unseen Split

We hold out one food type (for example, fruit) and split those dishes into two halves. The model is tested on the same dishes (Half 2) in two settings: after seeing this food in training, or never seeing it. Both training sets have the same size.

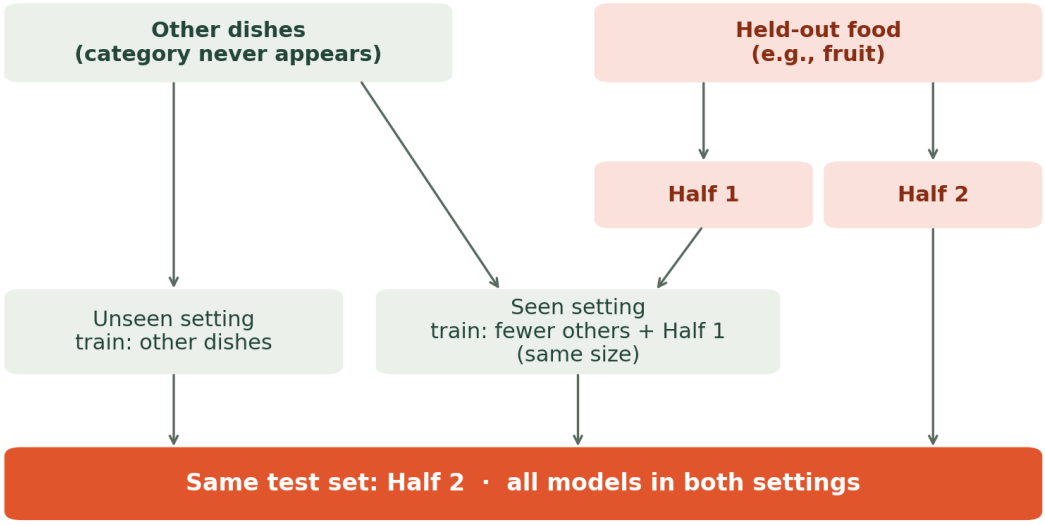


Figure 3. The only difference between the two settings is whether the model has seen this food.

5.3 Models

All models share the same general network backbone and training settings unless specified otherwise. Only the inputs and fusion mechanisms change. We use ResNet-50 as the primary baseline.

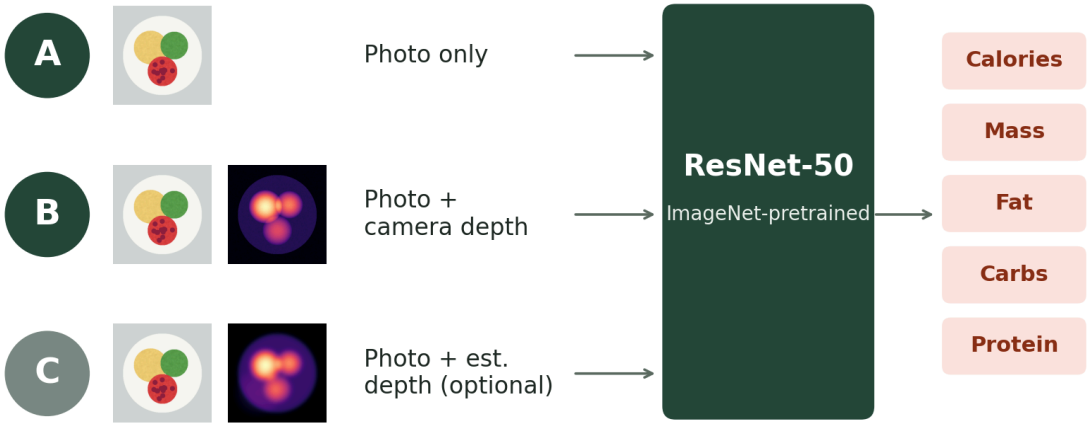


Figure 4. Models A, B, and C use the same ResNet-50 and predict the same five values. The RGB-D fusion network [2] and DPF-Nutrition [3] use dedicated fusion designs.

- **Phase 1:** Models A and B using ResNet-50. As an extension, we also evaluate an RGB-D fusion network [2] on the exact same seen/unseen split to see if dedicated multi-modal fusion improves generalization over standard concatenation.
- **Phase 2 (optional):** Model C using estimated depth from Depth Anything V2 [5], along with its extension utilizing DPF-Nutrition [3]. This phase will only be conducted if Phase 1 demonstrates a clear benefit from real depth and remaining time allows.

5.4 How We Measure

For each model, we measure how much its error grows on unseen foods (the "gap"). If model B's gap is clearly smaller than model A's, depth helps with new foods.

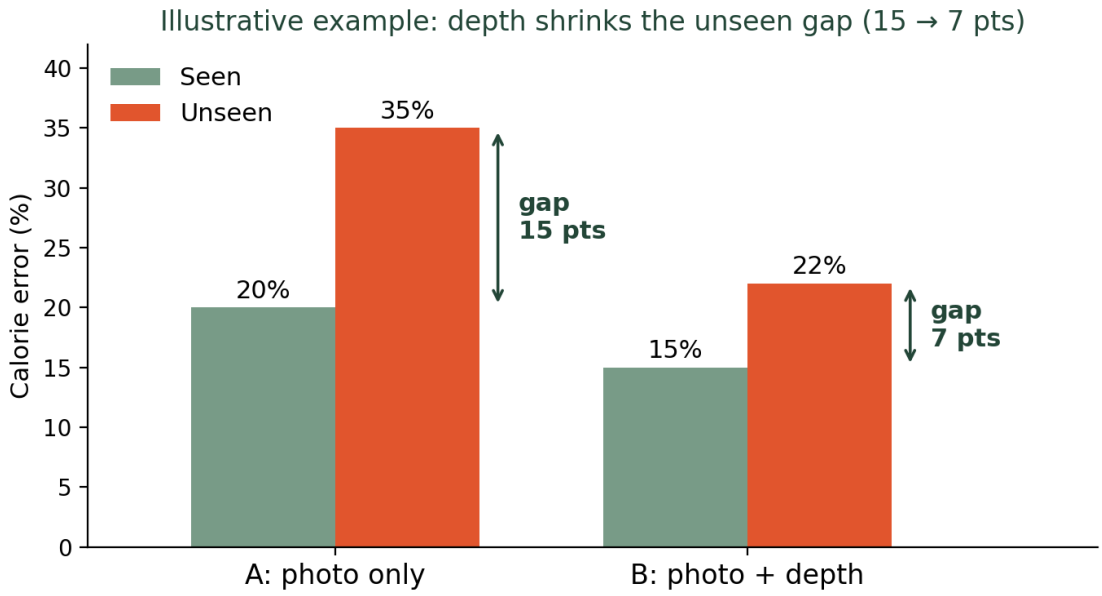


Figure 5. Illustrative example: depth reduces the unseen gap from 15 to 7 points.

We report mean absolute error with confidence intervals. To answer RQ2, we evaluate mass (portion) and calories per gram (food type) separately, using Nutrition5k's mass labels.

6. Expected Deliverables

- **Trained models:** A, B, and the RGB-D fusion network in both settings (C and DPF-Nutrition only if the optional Phase 2 is carried out).
- **Comparison results:** whether depth helps more on unseen foods.
- **Error analysis:** portion vs. food-type errors, with examples.
- **Report and code.** Stretch goal: a simple web demo.

7. Challenges

Challenge	Plan
A held-out food is often only part of a dish	Keep dishes where it is a major part
Small test set	Report confidence intervals honestly
Low-calorie foods inflate percentage error	Also report absolute error

8. Timeline

Week	Tasks
1	Data preparation and baseline
2	Phase 1: train models A, B, and the RGB-D fusion network
3	Analysis and final report (optional Phase 2 if time allows)

References

[1] Q. Thames et al., "Nutrition5k: Towards Automatic Nutritional Understanding of Generic Food," *CVPR*, 2021.

[2] For example, "RDINet: A Deep Learning Model Integrating RGB-D and Ingredient Features for Food Nutrition Estimation," *Applied Sciences*, 2026.

[3] Y. Han et al., "DPF-Nutrition: Food Nutrition Estimation via Depth Prediction and Fusion," *Foods*, 2023.

[4] For example, CalorieLLaVA, a fine-tuned LLaVA model for calorie estimation, 2024.

[5] L. Yang et al., "Depth Anything V2," *NeurIPS*, 2024.